HW2_31

2026-03-23

Data on weekly sales of a major brand of canned tuna by a supermarket chain in a large midwestern U.S. city during a mid-1990s calendar year are contained in the data file tuna. There are 52 observations for each of the variables. The variable sal1 = unit sales of brand no. 1 canned tuna, and apr1 = price per can of brand no. 1 tuna (in dollars).

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
tuna <- read.csv("tuna.csv")
tuna$WEEK <- 1:nrow(tuna)
str(tuna)
```

```
## 'data.frame':    52 obs. of  7 variables:
##  $ sal1  : int  6439 3329 3415 2909 2598 3773 20383 11761 2614 2496 ...
##  $ apr1  : num  0.66 0.62 0.62 0.62 0.63 0.69 0.63 0.65 0.77 0.81 ...
##  $ apr2  : num  0.82 0.8 0.77 0.66 0.65 0.63 0.65 0.78 0.62 0.69 ...
##  $ apr3  : num  0.79 0.59 0.63 0.81 0.81 0.8 0.78 0.8 0.77 0.68 ...
##  $ disp  : int  1 1 0 0 0 0 0 1 1 1 ...
##  $ dispad: int  0 0 0 0 0 0 1 0 0 0 ...
##  $ WEEK  : int  1 2 3 4 5 6 7 8 9 10 ...
```

(a) Summary Statistics and Time Plots

Calculate the summary statistics for sal1 and apr1. What are the sample means, minimum and maximum values, and their standard deviations. Plot each of these variables versus WEEK. How much variation in sales and price is there from week to week?

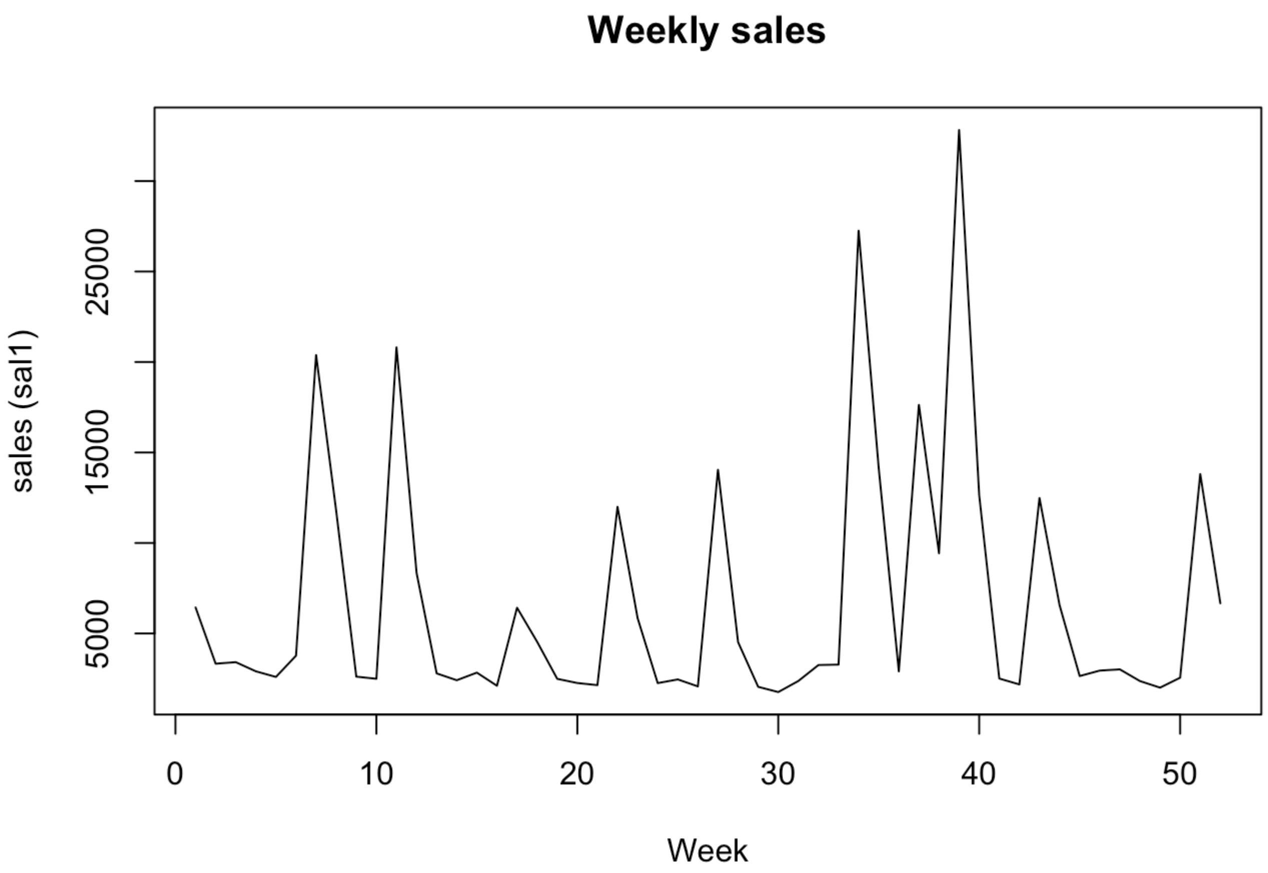
```
summary_stats <- tuna %>%
  summarise(
    mean_sal1 = mean(sal1),
    sd_sal1 = sd(sal1),
    min_sal1 = min(sal1),
    max_sal1 = max(sal1),
    mean_apr1 = mean(apr1),
    sd_apr1 = sd(apr1),
    min_apr1 = min(apr1),
    max_apr1 = max(apr1)
  )

summary_stats
```

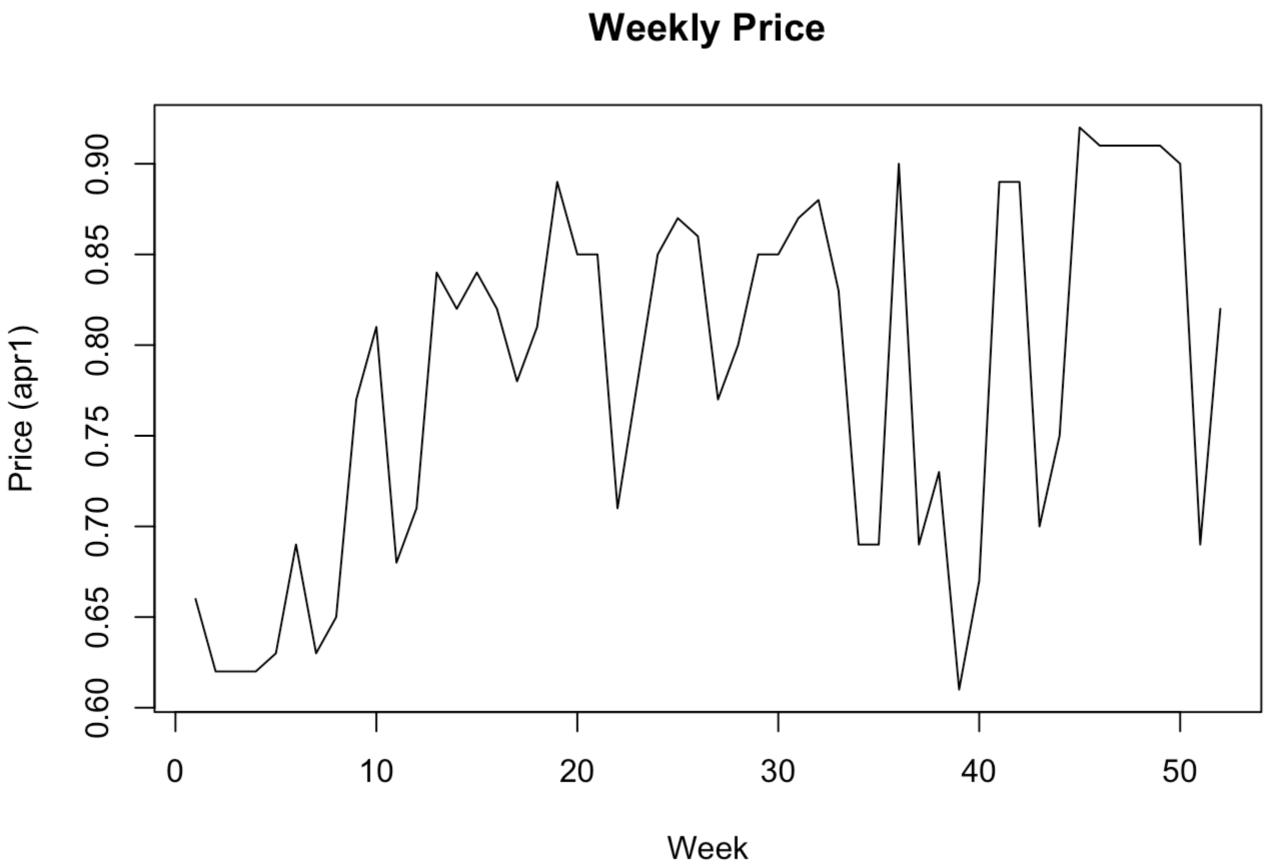
```
##   mean_sal1  sd_sal1 min_sal1 max_sal1 mean_apr1    sd_apr1 min_apr1 max_apr1
## 1  6718.712 6915.083     1762    32820    0.7825 0.09803711     0.61     0.92
```

Time Series Plots

```
plot(tuna$WEEK, tuna$sal1, type="l",
     xlab="Week", ylab="sales (sal1)",
     main="Weekly sales")
```




```
plot(tuna$WEEK, tuna$apr1, type="l",
     xlab="Week", ylab="Price (apr1)",
     main="Weekly Price")
```



*

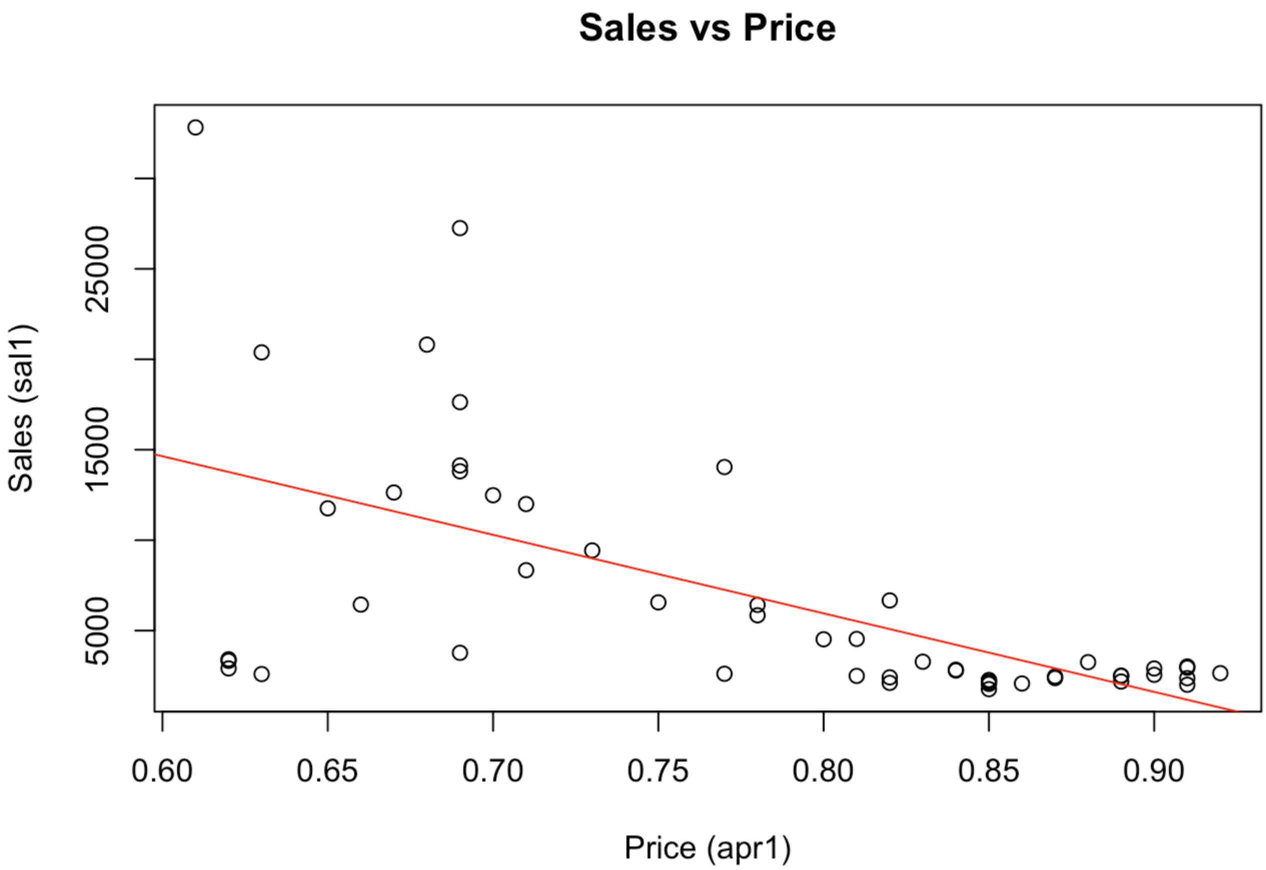
The price of tuna (apr1) shows relatively small variation over time, fluctuating within a narrow range without a clear trend. In contrast, sales (SAL1) exhibit substantial week-to-week variation, with several large spikes.

- These spikes in sales appear to coincide with periods of lower prices, consistent with the law of supply and demand, when prices decrease, sales volume increases.

(b) Scatter Plot: SAL1 vs apr1

Plot the variable SAL1 (y-axis) against APR1 (x-axis). Is there a positive or inverse relationship? Is that what you expected, or not? Why?

```
plot(tuna$apr1, tuna$sal1,
     xlab="Price (apr1)",
     ylab="Sales (sal1)",
     main="Sales vs Price")
abline(lm(sal1 ~ apr1, data=tuna), col="red")
```



There's an inverse relationship between two variables as expected, since it should be consistent with the law of supply and demand, when prices decrease, sales volume increases. ### (c) Regression with PRICE1 Create the variable $PRICE1 = 100APR1$. Estimate the linear regression $SAL1 = \beta_1 + \beta_2 PRICE1 + e$. What is the point estimate for the effect of a one cent increase in the price of brand no. 1 on the sales of brand no. 1? What is a 95% interval estimate for the effect of a one cent increase in the price of brand no. 1 on the sales of brand no. 1?

```
tuna <- tuna %>%
  mutate(PRICE1 = 100 * apr1)

model <- lm(sal1 ~ PRICE1, data = tuna)
summary(model)
```

```
##
## Call:
## lm(formula = sal1 ~ PRICE1, data = tuna)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10869.5  -1588.3   -499.3   1626.2  18607.1
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 40714.22    6196.42   6.571 2.82e-08 ***
## PRICE1       -434.45      78.58  -5.528 1.17e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5502 on 50 degrees of freedom
## Multiple R-squared:  0.3794, Adjusted R-squared:  0.367
## F-statistic: 30.56 on 1 and 50 DF,  p-value: 1.172e-06
```

- 95% Confidence Interval

$$\hat{\beta}_2 \pm t_{0.025,n-2} \cdot SE(\hat{\beta}_2)$$

```
beta2 <- coef(model)[2]
se_beta2 <- summary(model)$coefficients[2,2]
df <- nrow(tuna) - 2
t_crit <- qt(0.975, df)

cat("lower bound: ",beta2 - t_crit*se_beta2,"higher bound: ",beta2 + t_crit*se_beta2)
```

```
## lower bound:  -592.2897 higher bound:  -276.6049
```

- The estimated $\hat{\beta}_2$ is -434.45.
- In repeated sampling, about 95% intervals constructed this way will contain the true value.

(d) 90% CI for Expected Sales at Price = 70 cents

Construct a 90% interval estimate for the expected number of cans sold in a week when the price per can is 70 cents. * $70cents \rightarrow APR1 = 0.70, PRICE1 = 70$ * $E(SAL1|PRICE1 = 70) = \beta_1 + \beta_2 \cdot 70$

```
predict(model,
        newdata = data.frame(PRICE1 = 70),
        interval = "confidence",
        level = 0.90)
```

```
##       fit      lwr      upr
## 1 10302.9 8624.934 11980.87
```

- The estimated the expected number of cans sold is 10302.9.
- In repeated sampling, about 95% intervals constructed this way will contain the true value.

(e) Elasticity at the Means

Construct a 95% interval estimate of the elasticity of sales of brand no. 1 with respect to the price of brand no. 1 “at the means.” Treat the sample means as constants and not random variables. Do you find the sales are fairly elastic, or fairly inelastic, with respect to price? Does this make economic sense? Why?

$$E = \frac{\partial APR1}{\partial SAL1} \frac{SAL1}{APR1}$$

Since:

$$SAL1 = \beta_1 + \beta_2 (100APR1) \Rightarrow \frac{\partial APR1}{\partial SAL1} = 100\beta_2$$

Thus:

$$E = 100\beta_2 \frac{SAL1}{APR1}$$

```
APR1_bar <- mean(tuna$apr1)
SAL1_bar <- mean(tuna$sal1)

elasticity <- 100 * beta2 * (APR1_bar / SAL1_bar)
cat(elasticity)
```

```
## -5.059825
```

- 95% CI for Elasticity

$$SE(E) = 100 \frac{SAL1}{APR1} \cdot SE(\hat{\beta_2})$$

$$E \pm t_{0.025,n-2} \cdot SE(E)$$

```
se_E <- 100 * (APR1_bar / SAL1_bar) * se_beta2

cat("lower bound: ",elasticity - t_crit * se_E, "lower bound:",  elasticity + t_crit
* se_E)
```

```
## lower bound:  -6.898149 lower bound: -3.221501
```

- The estimated elasticity is -5.059825.
- In repeated sampling, about 95% intervals constructed this way will contain the true value.
- Since |E| > 1, demand is highly elastic. This implies that a 1% increase in price leads to more than a 5% decrease in quantity demanded, suggesting that consumers are very responsive to price changes, likely due to the availability of close substitutes and promotional pricing strategies.

(f) Hypothesis Test: Elasticity = -3

Test the hypothesis that elasticity of sales of brand no. 1 with respect to the price of brand no. 1 from part (e) is minus three against the alternative that the elasticity is not minus three. Use the 10% level of significance. Clearly, state the null and alternative hypotheses in terms of the model parameters, give the rejection region, and the p-value for the test. What is your conclusion?

- Hypotheses

$$H_0 : E = -3$$

$$H_1 : E \neq -3$$

- Test Statistic

$$t = \frac{E - (-3)}{SE(E)}$$

```
t_stat <- (elasticity + 3) / se_E
cat("t_stat:",t_stat,"\n")
```

```
## t_stat: -2.250572
```

```
t_crit_10 <- qt(0.95, df)
cat("t_critical:",t_crit_10,"\n")
```

```
## t_critical: 1.675905
```

```
p_value <- 2 * (1 - pt(abs(t_stat), df))
cat("p-value:",p_value)
```

```
## p-value: 0.02884204
```

- Rejection Region (α = 0.10): $|t| > t_{0.95,n-2}$. The absolute value of test statistic |t| = 2.250572 > t_critical = 1.675905.
- Since p-value = 0.02884204 < 0.10, we reject H_0 and conclude that the elasticity of sales of brand no. 1 with respect to the price of brand no. 1 is not -3.